

gdTAI V4.1-GPU Gate B QC

GPU implementation readiness without fitting any project cell.

PASS. The frozen cache contract, deterministic A100 sentinels, portable model reloads, threshold frontiers, and atomic checkpoint resume all passed. Project fitting remains locked behind Gate C.

Scope

Included: implementation, synthetic GPU fitting, serialization, checkpointing, and read-only validation.

Excluded: project-data fitting, model selection, promotion, release fitting, and atlas inference.

Compute Contract

cuda_device_name	NVIDIA A100 80GB PCIe
cuda_device_count	1
torch_version	2.9.1+cu128
xgboost_version	2.1.2
cuda_visible_devices	0
cuda_mps_pipe_directory	/tmp/gdtai-v41-gateb-20260807-direct
cublas_workspace_config	:4096:8
pythonhashseed	20260807

Validation Results

Check	Status	Evidence
preflight_pass	PASS	PASS
cell_manifest_sha256	PASS	8157cbebfedeb84fc34eba05429aa8a7a834c6f7ceba15bd4790b5dd06bf7e0c
split_manifest_sha256	PASS	c84da2ca999676bab0ed180ae3d380e1e7d5b2e08da886ae2f6f912f9e0080a7
feature_manifest_sha256	PASS	7fc2b0afb2b761288f941895a219fb38cc541a49464c8d2b330b81719e087808
feature_count	PASS	197
feature_unique	PASS	duplicates=0
cell_count	PASS	1137739
cell_ids_unique	PASS	duplicates=0
sensitivity_excluded_from_fit	PASS	violations=0
cache_cell_manifest	PASS	8157cbebfedeb84fc34eba05429aa8a7a834c6f7ceba15bd4790b5dd06bf7e0c
cache_feature_manifest	PASS	7fc2b0afb2b761288f941895a219fb38cc541a49464c8d2b330b81719e087808
cache_shape	PASS	(1137739, 197, 8)

Check	Status	Evidence
cache_exclusion_audit	PASS	{'all_cell_cd4_helper_excluded': 167270, 'all_cell_treg_excluded': 36350, 'all_cell_union_excluded': 186244, 'preflight_scope_cells': 397814, 'preflight_scope_reproduction_pass': True}
gene_feature_shape_dtype	PASS	shape=(1137739, 197); dtype=float32
gene_feature_sha256	PASS	97b2a719cfe77526005941fce289e676518513cec7104aa7770825f5c92c4350
derived_feature_shape_dtype	PASS	shape=(1137739, 8); dtype=float32
derived_feature_sha256	PASS	8c24b0a3ad3e4a713108aa1f24a5929fc89ee0550d725c368f1e73b1859ed41e
legacy_score_shape_dtype	PASS	shape=(1137739,); dtype=float32
legacy_score_sha256	PASS	a5f5f4564bc1b50c67d0099aabe4c947ca84bf695ee03738796010d487b51164
torch_ridge_bit_identical	PASS	max_abs=0
torch_ridge_finite	PASS	n=8192
torch_ridge_convergence	PASS	PASS
torch_ridge_portable_export	PASS	{ "metadata": "aded1aaea9f10170e3c0d13dd975630db4a431174675f4b03880423ea237ee04", "model": "4572123049db6163f41f9592e613d9f0ceceb173c16804874af394b0a40f3cd4" }
torch_ridge_portable_reload	PASS	max_abs=0
xgboost_bit_identical	PASS	max_abs=0
xgboost_finite	PASS	n=8192
xgboost_convergence	PASS	PASS
xgboost_portable_export	PASS	{ "metadata": "a2133a639430f64f8f85c2935985fb1ba11e987de198966c41b42888e9e5bfa9", "model": "1bd7049b73ed1e9db3138ad4ead79eb034d96c30be39d5b2abf20afa8c95a86a" }
xgboost_portable_reload	PASS	max_abs=0
stage1_complete_frontier	PASS	{ "candidate_count": 18, "candidate_sha256": "777a1b88d78dc7bbcl808ddfab1f5641af844e3fa30b7f24a6f38d0d2daec61", "macro_f0_5": null, "macro_f1": null, "macro_precision": null, "macro_recall": 1.0, "mode": "stage1", "objective_value": 1.0, "paired_abt_fpr": null, "passed": true, "reason": "PASS", "strict_nk_fpr": 0.0, "threshold": 0.94, "valid_candidate_count": 7 }
stage2_complete_frontier	PASS	{ "candidate_count": 11, "candidate_sha256": "5b4eb0f76084e92ed881f17794d0e0659d92cb8f0328ff52b49a63551a91c0f4", "macro_f0_5": 1.0, "macro_f1": 1.0, "macro_precision": 1.0, "macro_recall": 1.0, "mode": "synthetic", "objective_value": 1.0, "paired_abt_fpr": 0.0, "passed": true, "reason": "PASS", "strict_nk_fpr": 0.0, "threshold": 0.7, "valid_candidate_count": 1 }
atomic_checkpoint_resume	PASS	/tmp/gdtai-v41-gateb-o0l9wz9g/checkpoints/synthetic

Next Supervision Action

Review this package. A separate, checksum-bound Gate C approval is required before any nested project-data fit can start. No guardrail or feature rule has been changed.